

Johansen Cointegration Test

Date: 07/08/26 Time: 11:22
Sample: 1/04/2010 12/30/2025
Included observations: 4159
Lags interval (in first differences): 1 to 4
Endogenous variables: EURUSD GBPUSD
Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None *	0.002651	16.46928	15.49471	0.0356
At most 1 *	0.001309	5.441322	3.841465	0.0197

Trace test indicates 2 cointegrating equation(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.002651	11.02796	14.26460	0.1528
At most 1 *	0.001309	5.441322	3.841465	0.0197

Max-eigenvalue test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegrating Coefficients (norma

EURUSD	GBPUSD
17.33760	-10.44034
-1.981938	7.983808

Unrestricted Adjustment Coefficients (alpha):

D(EURUSD)	-0.000134	-0.000214
D(GBPUSD)	0.000185	-0.000243

1 Cointegrating Equation
Log-Likelihood: 30436.42

Normalized cointegrating coeffici

EURUSD	GBPUSD
1.000000	-0.602179
	(0.11724)

Adjustment coefficients (standard

D(EURUSD)	-0.002315	(0.00174)
D(GBPUSD)	0.003201	(0.00205)